

Referee report, 2nd round, “reviewer 2”

The author has adequately addressed many, especially the major points from the first referee report. In the author’s response, the rejection of other points is reasonably justified in some cases, less convincingly in others. Even in the latter case, this is acceptable in many instances.

Rather than further expanding on policy issues, the author has reduced the ambition for policy relevance. This sharpens the paper’s focus and makes some of the omissions more acceptable. Then again, it also makes the paper less policy relevant. It becomes apparent that the second part of the title (“the distribution of impacts of Swiss climate policy”) is only half accurate, given the stylized nature of the exercise. This stylized nature is already described. It should be made more transparent with respect to the issue of neglected abatement costs (see below).

In the following I concentrate on the few places which I am still suggesting changes for. With these minor adjustments, I recommend the paper for publication.

- General parts of the 1st referee report
 - o The additions that point to the short-term nature of the analysis are fine. In the introduction, line 217 is the ideal place to also mention it.
 - o The additions on feasibility are also fine. Indeed, the first contribution has been wisely reformulated as well. In line 1803 I am still stumbling over the words “desirable” and “feasible”. How about “viable” to combine the two?
- Line 193ff (now line 190ff): Considering the focus and main recommendations of the paper, the difficulty in targeting the emissions of the rich cannot be ignored. Not only are they less sensitive to price changes. The rich are also able to lobby for exemptions, and the emissions of the very rich grow faster than anyone else’s. It’s not merely a fairness issue, but also concerns the effectiveness of policy measures. Even if the author has chosen to merely “target goods” and discuss fairness, it is a caveat to be mentioned that the proposed prescriptions may partially fail to effectively reduce emissions.
- Line 224ff (now line 220ff): That “technological change sweeps through the energy sector” could be mentioned not only in the response, but also in the paper.
- Line 1344 (now line 1249f): In a chapter entitled “Climate policy design” there should be space for mentioning this. It only takes half a sentence. As a minimum requirement, the current sentence must be modified, because “without affecting the consumption of low-income households too much” is incorrect in the perception of those facing the regulation. Replacing “consumption” with “cost of living” would do.
- Line 1580 (now line 1488ff): It is impossible for a modeler not to make “any assumptions with respect to preexisting taxes such as mineral oil taxes”. The prices used are either net or gross of taxes before levying the CO2 tax. With constant demands, the influence on results is indeed smaller than in e.g. a CGE analysis. Given the large magnitude of the mineral oil tax, the distributional results are, however, not unaffected by the assumption of taking into account – or not taking into account – the mineral oil tax. Hence, the

approach must be briefly described (i.e. do the prices net of carbon taxes include the mineral oil tax or not?) and motivated.

- Full redistribution of CO2 tax revenues: This must be transparently explained, and the associated caveat of neglecting the costs of emission reductions must be prominently stated. So far, the paper mentions costly emission reductions in many instances, giving the misleading impression that this is what the paper is about. It is really about distributional effects related purely to the payments that directly belong to the policy instrument. Correcting this in the text requires modifications in many locations. This is a non-exhaustive list of locations with references to costly emission reductions that will have to be presented differently:
 - Line 198: measuring “exposure to costly climate policy”
 - Line 212f: “Second, the paper discusses exposure of different income classes to costly climate policy.” -> especially important to change this formulation, as it describes one of the main objectives of the paper
 - Line 715: “costly reduction of the emissions”
 - Line 1002f: “requiring costly emission reduction”
 - Line 1010f: “higher costs from reducing emissions”
 - Line 1133: “costly climate policy”
 - Line 2011ff: “I assume that emission free technologies for producing different consumption goods are more expensive than their polluting counterparts at least in the short and medium run and that the state will not fully take over these additional costs.” (further explained in footnote 17) This is incompatible with the assumed full redistribution of revenues.
 - Line 1337f: “overall costs of achieving emissions reduction”
 - Line 1370: “The consequence of costly emissions reduction”
 - Figure 10 shows welfare costs
 - Line 1735: “costly emissions reduction efforts”
- Line 1636ff (now line 1546): In this case I suggest adding “and communicated” after “properly designed”. This is necessary also because the sentence itself refers to the “perception of regressivity”.
- Appendix B: The title should be changed to “Household types versus income as determinants”. This helps the reader to immediately grasp what it is about.